AI-BASED FOREST COVER CHANGE DETECTION SYSTEM

Milestone-1
(Group-5)

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Project Overview & Objectives

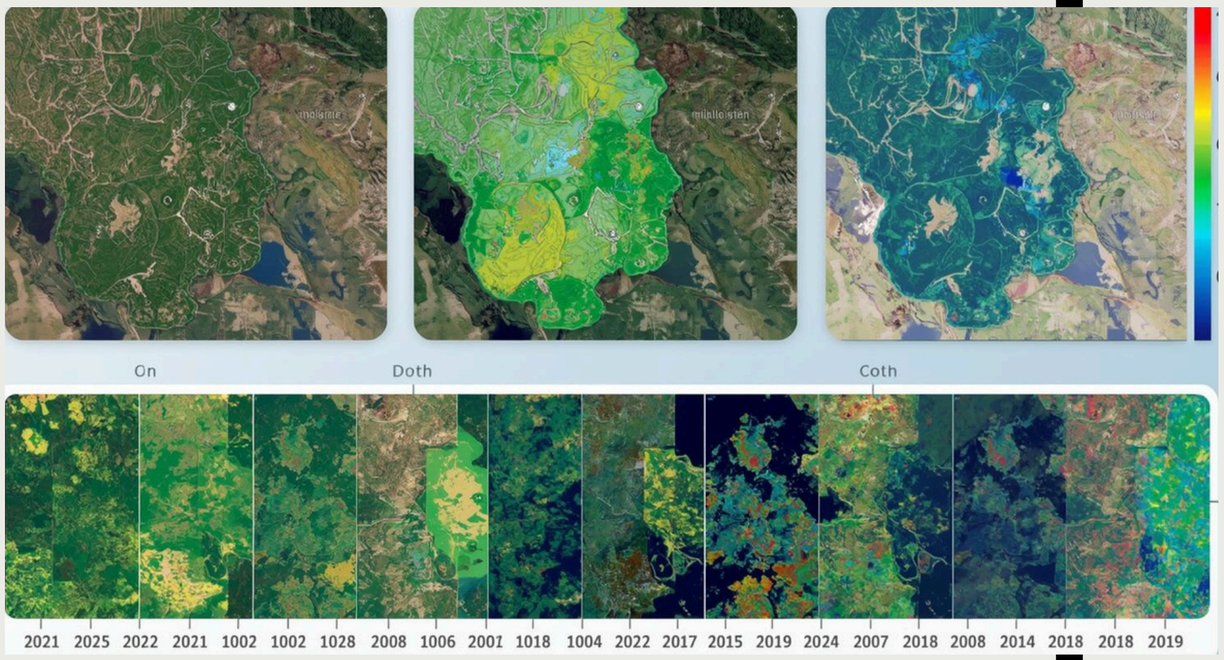
Primary Objective: Develop specialized AI-based monitoring system using multi-temporal Sentinel satellite imagery at 10-meter resolution with deep learning semantic segmentation over a 5-year period.

Target Regions:

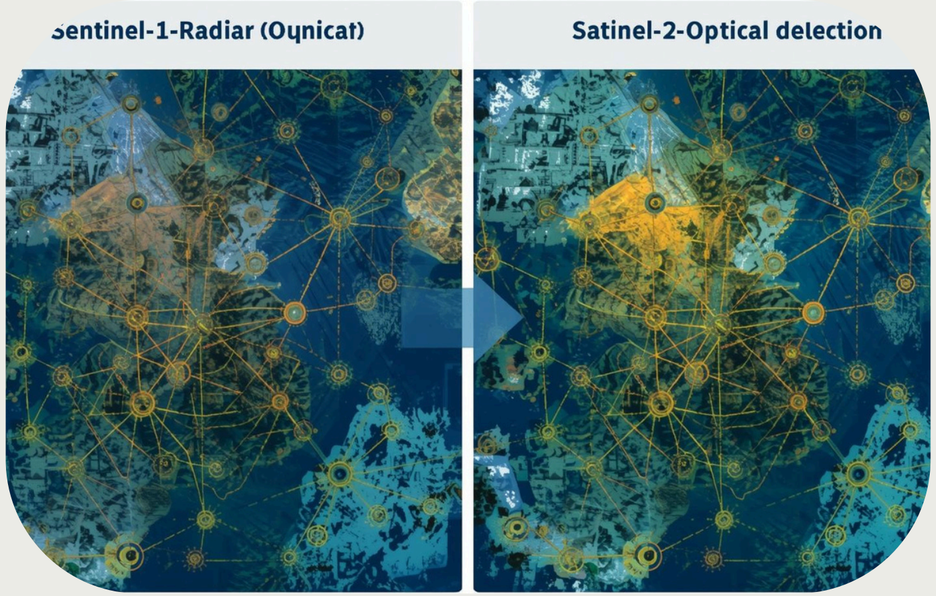
- Aravalli Range: 692 km natural barrier against Thar Desert
- North Eastern Region (NER): Global biodiversity hotspot with dense tropical forests

Key Threats:

- Aravallis: Unregulated mining and rapid urbanization
- NER: Urban sprawl and traditional shifting cultivation (Jhum)



TECHNICAL APPROACH & INNOVATION



Core Technology:

- Siamese U-Net Architecture for direct change feature learning
- Multi-modal Data Fusion: Sentinel-1 (Radar) + Sentinel-2 (Optical)
- 10-meter spatial resolution for precise detection

Key Innovations:

- Pretrained segmentation backbones with region-specific fine-tuning
- Automated change mapping framework
- Cloud-robust monitoring using radar integration
- Detection of small-scale deforestation activities (Jhum cultivation)

ADDRESSING CURRENT LIMITATIONS

Identified Gaps in Existing Solutions

Challenge	Current Solutions	Our Approach
Cloud Coverage	Hansen GFC: "Holes" during monsoon	Radar + Optical fusion
Resolution	30m (misses small Jhum patches)	10m high-resolution detection
Temporal Lag	1-2 year reporting delay	Near real-time monitoring
Change Detection	Post-classification comparison	Direct feature-level learning

Current Baselines:

- Hansen GFC: ~75% accuracy in fragmented forests
- Dynamic World: ~0.72 mIoU for tree cover

PERFORMANCE TARGETS & METHODOLOGY

Improvement Goals & Strategy

Target Metrics:

- F1-Score: > 0.80 (vs. current 0.70-0.75)
- IoU (Intersection over Union): > 0.70

Three-Pillar Improvement Strategy:

1. Radar Fusion: Eliminate cloud-caused data gaps
2. Siamese Learning: Reduce noise from separate classifications
3. Spatial Refinement: Capture narrow logging roads and small patches

Data Sources:

- Sentinel-2 (L2A): 10m multi-spectral bands for vegetation health
- Sentinel-1 (GRD): C-band radar for structural changes
- Hansen GFC (v1.11): Supervised training labels

EVALUATION FRAMEWORK & EXPECTED IMPACT

Comprehensive Validation Strategy

Three-Tier Evaluation:

1. Quantitative: Precision, Recall, F1-Score against Hansen test set
2. Cross-Product Comparison: Direct comparison with Dynamic World subtraction maps
3. Visual Ground Truth: Manual verification using Google Earth Pro high-resolution imagery

Key Differentiator:

- Spatial Context Analysis: Unlike existing pixel-based solutions, our Siamese U-Net analyzes texture and patterns to distinguish permanent deforestation from seasonal agricultural changes

Expected Impact:

- Enhanced monitoring of ecologically sensitive regions
- Improved detection of small-scale forest fragmentation
- Real-time intervention capabilities for forest conservation

THANK YOU

SO MUCH!